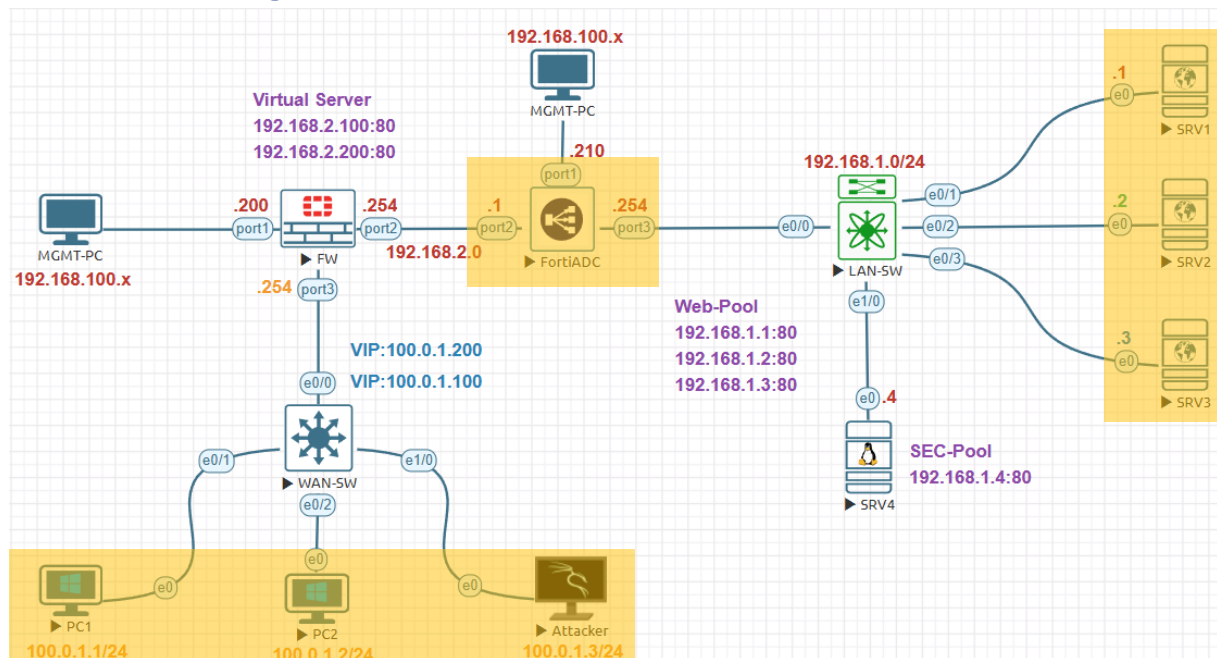


Content Routing Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Create Resources in SRV3:

First let's create resources in **SRV3** only, which does not exist on the other two Servers.

Login to SRV3 with username/password root/root

```
root@SRV3:~# cd /var/www/html/  
root@SRV3:/var/www/html# vi rs.html
```

press I to enter insert mode type below

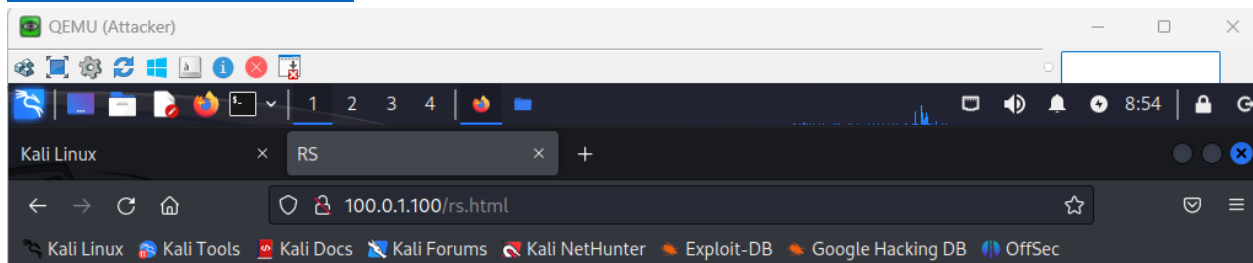
```
<html>  
<head>  
<Title>RS</title>  
<body>  
<h1>  
This Test Resources which is only available in SRV3.  
</h1>  
</body>  
</html>
```

Press ESC to exit from writing mode and type :wq

Testing:

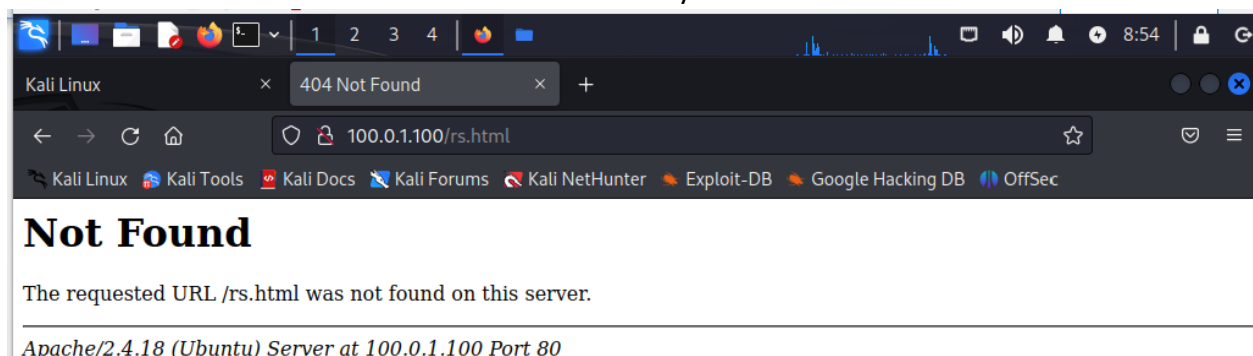
On the Kali Linux External, open a browser, and then connect to the following URL:

<http://100.0.1.100/rs.html>



This Test Resources which is only available in SRV3.

Press **Ctrl+Shift+R** to refresh the browser page several times. Approximately two out of three connection attempts will fail. They fail each time the device routes traffic to SRV1 and SRV2 because the resources folder is located on SRV3 only.



Create New Real Server Pool:

Click **Server Load Balance > Real Server Pool > Real Server Pool**.

Click **Create New** to create a new real server pool using the following settings:

Name: **SRV3-Pool**

Address Type: **IPv4**

Health Check: **On**

In the Available Items list, double-click **HTTP_Check** to add it to the Selected Items list.

Keep the default values for the remaining settings, and then click **Save**.

Dashboard > Security Fabric > FortiView > System > Shared Resources > Network > **Server Load Balance** > **Real Server Pool** > Scripting > SSL-FP Resources > Link Load Balance > Global Load Balance > Web Application Firewall > Network Security > DoS Protection > User Authentication > Log & Report

Real Server Pool

Name: SRV3-Pool

Address Type: IPv4 IPv6

Type: Static Dynamic

Health Check: ☒

Health Check Relationship: AND OR

Selected Items: LB_HLTHCK_HTTP

Available Items: Create New, LB_HLTHCK_ICMP, LB_HLTHCK_HTTPS, LB_HLTHCK_TCP_ECHO

Health Check List: Double-click to deselect. Drag to reorder.

Direct Route Mode: ☐

Real Server SSL Profile: NONE

Member

ID	Name	Address	Health
No data available			

Showing 0 to 0 of 0 entries 0 rows selected Show 25 entries

Save

Finally, we have new Real Server Pool with the name **SRV3-Pool**.

Name	Address Type	Health Check
WEB-Pool	IPv4	Enable
SEC-Pool	IPv4	Disable
SRV3-Pool	IPv4	Enable

Add Members to Real Server Pool:

Click the edit icon or **double click** to **edit** the new real server pool named **SRV3-Pool**.

Real Server Pool Real Server Server SSL		
Delete + Create New Add Filter		
Name	Address Type	Health Check
WEB-Pool	IPv4	Enable
SEC-Pool	IPv4	Disable
SRV3-Pool	IPv4	Enable

Scroll down to the **Member** pane and click **Create New** to create a new member using the following settings:

Status: **Enable**

Real Server: **SRV3**

Port: **80**

Weight: **1**

Keep the default values for the remaining settings, click **Save**, and then click **Save** again.

Real Server Pool	Member
Name	Status Enable Disable Maintain
Address Type	Real Server SRV3
Type	Port 80 Default: 80 Range: 0-65535
Health Check	Weight 1 Default: 1 Range: 1-256
Health Check Relative	Recover 0 Default: 0 (disabled) Range: 0-86400 seconds
Health Check List	Warm Up 0 Default: 0 (disabled) Range: 0-86400 seconds
Direct Route Mode	Warm Rate 100
Modify Host	☐
Save Cancel	

Real Server Pool

Name:

Address Type: ☒ IPv4 ☐ IPv6

Type: ☒ Static ☐ Dynamic

Health Check: ☒

Health Check Relationship: ☒ AND ☐ OR

Health Check List:

Selected Items

- LB_HLTHCK_HTTP

Double-click to deselect. Drag to reorder.

Available Items

- Create New
- LB_HLTHCK_JCMP
- LB_HLTHCK_HTTPS
- LB_HLTHCK_TCP_ECHO

Double-click to select.

Direct Route Mode: ☐

Real Server SSL Profile:

Member

ID	Name	Address	Health Check	Port
1	SRV3	192.168.1.3	inherited	80

Showing 1 to 1 of 1 entries 0 rows selected Show entries

Create Content Routes:

Click **Server Load Balance > Virtual Server > Content Routing**.

Click **Create New** to create a new route using the following settings:

☒ Dashboard >
 ☒ Security Fabric >
 ☒ FortiView >
 ☒ System >
 ☒ Shared Resources >
 ☒ Network >
 ☒ **Server Load Balance** >

☒ Virtual Server
 ☒ Content Rewriting
 ☒ **Content Routing**

Name	Type	Real Server
------	------	-------------

Showing 0 to 0 of 0 entries 0 rows selected Show entries

Virtual Server

Name: SRV3-RS-Route

Type: Layer 7

Real Server Pool: SRV3-Pool

Persistence: Inherit

Method: Inherit

Click **Save** and then click **Save** again.

Dashboard > Security Fabric > FortiView > System > Shared Resources > Network > **Server Load Balance** > **Virtual Server**

Application Resources
Application Optimization
Real Server Pool

Content Routing

Name: SRV3-RS-Route

Type: Layer 4 Layer 7

General

Schedule Pool: ☐

Real Server Pool: SRV3-Pool

Persistence: Click to select

Inherit: ☒

Method: LB_METHOD_ROUND_ROBIN

Inherit: ☒

Click **Server Load Balance > Virtual Server > Content Routing**.

Click **Create New** to create a new route using the following settings:

Dashboard > Security Fabric > FortiView > System > Shared Resources > Network > **Server Load Balance** > **Virtual Server**

Application Resources
Application Optimization
Real Server Pool

Virtual Server Content Rewriting **Content Routing** N.

Delete **+ Create New** + Add Filter

Name	Type	Real Server
------	------	-------------

Showing 0 to 0 of 0 entries 0 rows selected Show 25 entries

Name: All

Type: Layer 7

Real Server Pool: WEB-Pool

Persistence: Inherit

Method: Inherit

Click **Save** and then click **Save** again.

Dashboard

Security Fabric

FortiView

System

Shared Resources

Network

Server Load Balance

Virtual Server

Application Resources

Application Optimization

Real Server Pool

Scripting

Content Routing

Name

All

Type

Layer 4

Layer 7

General

Schedule Pool

Real Server Pool

WEB-Pool

Persistence

Click to select

Inherit

Method

LB_METHOD_ROUND_ROBIN

Inherit

Comments

Finally, we have now two content routing rules, as follows.

Dashboard	Virtual Server	Content Rewriting	Content Routing	NAT Source Pool
Security Fabric	Delete	Create New	Add Filter	
FortiView	Name	Type	Real Server	
System	SRV3-RS-Route	Layer 7	SRV3-Pool	
Shared Resources	All	Layer 7	WEB-Pool	
Network	Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries			
Server Load Balance				
Virtual Server				

Match Conditions to Content Routes:

Click the **edit** icon or double click to edit the **SRV3-RS-Route**.

The screenshot shows the FortiGate web interface with the 'Content Routing' tab selected. The left sidebar shows the navigation menu with 'Server Load Balance' expanded. The main area displays a table of content routes:

Name	Type	Real Server
SRV3-RS-Route	Layer 7	SRV3-Pool
All	Layer 7	WEB-Pool

Below the table, it says 'Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries'. The 'Virtual Server' tab is also visible at the bottom.

Scroll down to the **Match Condition** pane, and then click **Create New** to create a new rule using the following settings:

Object: HTTP Request URL

Type: String

Content: rs.html

The screenshot shows the 'Match Condition' configuration pane for the 'SRV3-RS-Route'. The settings are as follows:

- Name: SRV3-RS-Route
- Type: Layer 4
- Match Condition (Empty Match Condition will match anything)
- Object: HTTP Request URL
- Type: String (selected), Regular Expression
- Content: rs.html
- Negative: ☐
- Ignore Case: ☒
- Buttons: Save, Cancel

Click **Save**, and then click **Save** again.

The screenshot shows the 'Match Condition' configuration pane after saving. The settings are as follows:

- Match Condition (Empty Match Condition will match anything)
- Buttons: Delete, Create New, Add Filter
- Table:

ID	Object	Type	Content
1	HTTP Request URL	String	rs.html

Below the table, it says 'Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries'. The 'Save' button is highlighted.

Click the **edit** icon or double click to edit the **All** content route.

Virtual Server Content Rewriting Content Routing NAT Source Pool Schedule Pool			
Delete + Create New Add Filter			
Name	Type	Real Server	Match C
SRV3-RS-Route	Layer 7	SRV3-Pool	1
All	Layer 7	WEB-Pool	0
Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries			

Scroll down to the **Match Condition** pane, and then click **Create New** to create a new rule using the following settings:

Object: HTTP Host Header

Type: String

Content: 100.0.1.100

Click **Save**, and then click **Save** again.

Content Routing

NameAll

TypeLayer 4

General

Schedule Pool

Real Server Pool

Persistence

Inherit

Match Condition (Empty Match Condition will match anything)

ObjectHTTP Host Header

TypeStringRegular Expression

Content100.0.1.100

Negative

Ignore Case

Save

Cancel

Match Condition (Empty Match Condition will match anything)

Delete + Create New Add Filter			
ID	Object	Type	Content
1	HTTP Host Header	String	100.0.1.100
Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries			

Save

Cancel

Enable Content Routes:

Click **Server Load Balance > Virtual Server > Virtual Server**. Click the **edit** icon or double click to edit **WEB_VS**.

Name	Type	Address	Port
WEB-VS	Layer 7	192.168.2.100	80
SEC-VS	Layer 7	192.168.2.200	80

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

In the **Specifics** pane, enable **Content Routing**.

Content Routing ☒

Content Routing List

Selected Items

- SRV3-RS-Route
- All

Available Items

- Create New
- SRV3-RS-Route
- All

Double-click to deselect. Drag to reorder.

Double-click to select.

In the **Available Items** list, double-click **SRV3-RS-Route** to add it to the **Selected Items** list. In the **Available Items** list, double-click **All** to add it to the **Selected Items** list. The route **All** must be the second entry in the **Selected Items** list from top to bottom.

Content Routing ☒

Content Routing List

Selected Items

- SRV3-RS-Route
- All

Available Items

- Create New
- SRV3-RS-Route
- All

Double-click to deselect. Drag to reorder.

Double-click to select.

If necessary, you can drag and drop the content routes to reorder them.

Click the **General** tab.

The screenshot shows the FortiGate configuration interface. On the left is a sidebar menu with items: Dashboard, Security Fabric, FortiView, System, Shared Resources, Network, and **Server Load Balance** (highlighted with a red box). Below the menu is a 'Virtual Server' button. The main area is titled 'Virtual Server' and has four tabs: Basic, **General** (highlighted with a red box), Security, and Application Optimization. Under the 'General' tab, there is a 'Configuration' section with two fields: 'Address' with the value '192.168.2.100' and 'Port' with the value '80'. Below the 'Port' field is a note: 'Default: 80 Range: 0 or 1-65535. You can specify up to eig e.g., 80-90 100. Valid values are from 0 to 65535, with 0 fo'.

Scroll down to the **Error Page** pane, and then type the following message in the **Error Message** field to customize it:

Server is not currently available. Please try again later.

Click **Save**.

The screenshot shows the 'Error Page' pane in the FortiGate configuration interface. The 'Error Page' dropdown menu is set to 'Click to select'. The 'Error Message' text area contains the message 'Server is not currently available. Please try again later.' (highlighted with a red box). Below this is the 'FortiGSLB' section with three options: 'Public IP Type' with 'IPv4' selected, 'Public IPv4' with the value '0.0.0.0' and an example '192.0.2.1', and 'One Click GSLB Server' which is a toggle switch currently turned off. At the bottom right, there is a red arrow pointing to a green 'Save' button.